An integrated scRNA-seq lung atlas built from agent-curated datasets (TODO: Be ”cell” vs ”barcode”)

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Abstract

The abundance of nucleotide sequence data has grown into the order of trillions of bases. As public archives grow in scale, so too does their fragmentation and complexity, meaning that disease-focused meta-analyses remain difficult to accomplish. We identified the need for producing atlas-scale datasets that both incorporate work spanning studies and are uniformly processed and tissue-targeted. scBaseCount hosts uniformly processed single-cell RNA sequencing files, but those files are not merged, batch corrected, or clustered. To address these shortcomings, we selected datasets whose scBaseCount free-text tissue metadata matched lung, applied quality control, and combined them into one atlas. Concatenation exposed a study-level batch effect, but scBaseCount contains no batch variable coarser than SRA experiment accession. We therefore retrieved National Center for Biotechnology Information's BioProject accessions from the European Nucleotide Archive and used them as the Harmony batch key. Finally, we clustered the neighborhood graph across a range of resolutions and selected the resolution that maximized the summed Jaccard index of the optimal one-to-one matching between clusters and published cell-type labels. The atlas contains 9,307,963 barcodes from 1,410 SRA and ENA experiment accessions spanning 172 BioProject accessions. Follow-up checks showed that cells from different studies mixed

more after batch correction, while cell-type structure was largely retained. The scale and integration of the clustered dataset presents a possibility of making novel cross-study findings, even across rare cell types, through downstream analysis.

Introduction

Public single-cell RNA sequencing (scRNA-seq) archives now contain enough cells for tissue-scale reuse, but experiments are still deposited independently, processed with different pipelines, and described with free-text metadata (Luecken et al., 2022; Mueller et al., 2026). Cross-study analysis therefore rarely starts from a shared gene axis or a study-level batch structure. Building an atlas from public data remains a separate construction problem, whereby a tissue must be selected, matrices must be brought onto a common reference, and the joint object must be integrated without erasing cell-type biology.

Community contributed collections such as CZ CELLxGENE Discover enforce cell-level and metadata standardization on submission, allowing data exploration and meta-analysis (CZI Cell Science Program et al., 2025). Efforts like the Human Lung Cell Atlas (a part of the Human Cell Atlas project) integrate multiple selected respiratory studies into a consensus cell-type map (Regev et al., 2017; Sikkema et al., 2023). These resources straddle the inevitable tradeoff between being exhaustive and curated. In the case of the mentioned efforts they do not ingest the Sequence Read Archive (SRA) as a whole (Leinonen et al., 2011). On the other hand, scBaseCount opts for extensiveness by using an agentic AI metadata curation approach to collect and reprocess public 10X scRNA-seq experiments at large scale, producing a library of `h5ad` files keyed by SRA experiment accession (Youngblut et al., 2025). Those files do not contain usable clusters. They are also neither tissue-restricted nor treated with any batch correction.

Here we construct an integrated lung-tissue atlas from the `2026-01-12` scBaseCount human GeneFull release. We focus on lung tissue because of the frequency and severity of lung diseases. Lung cancer was estimated to be the most frequently diagnosed cancer at 2.6 million new cases, and the leading cause of cancer death with 1.9 million deaths in 2024 (Sung et al., 2026). Non-cancer diseases such as idiopathic pulmonary fibrosis are also a cause for concern and are known to be associated with high mortality rates (Chakraborty et al., 2022). scRNA-seq has promise and meaningful progess when it comes to researching lung diseases such as these (Chakraborty et al., 2022; Navin et al., 2011). To focus on lung tissue samples, we select experiments from scBaseCount whose tissue labels match lung. We apply per-dataset quality control, align datasets by gene and concatenate them over cell them into a single object. We determine that scBaseCount does not contain any cross-dataset variable that could serve as a key to correct for the batch effect. Therefore, we retrieve the National Center for Biotechnology Information's (NCBI) BioProject accessions for each dataset and use them as the Harmony batch variable (Barrett et al., 2012; Korsunsky et al., 2019). We also develop a method to optimize a clustering algorithm parameter, namely the Leiden algorithm's resolution parameter, by matching clusters to CZ CELLxGENE `cell_type` labels (Traag et al., 2019). Finally, we evaluate batch correction with a suite of metrics proposed by Luecken et al. (2022) on a study-stratified subset. The result is an analysis-ready lung atlas derived from an agent-curated public resource.

Materials and methods

Building an integrated lung atlas using scBaseCount data required developing robust single-cell workflows in Python using Scanpy (Wolf et al., 2018). We enforced observability through extensive logging configured with the Python logger.

Data acquisition

scBaseCount data, which includes expression matrices, are hosted on Google Cloud Platform in Google Cloud Storage (GCS) buckets (Youngblut et al., 2025). We used human, GeneFull matrices from the `2026-01-12` release of scBaseCount. Before downloading any matrices, we identified a subset of the available matrices from corresponding metadata table shipped with scBaseCount. We discarded accessions that were not SRA or ENA records (identifiers beginning with NRX). We then flagged lung tissue by applying a regular expression to the free-text tissue field, matching terms such as "lung," "pulmonary," "alveolus," and "bronchus." This selection used the scBaseCount metadata string and did not independently verify the tissue represented by each accession. For each remaining accession we retrieved a BioProject accession from the European Nucleotide Archive (ENA) portal API (Amid et al., 2020). We excluded seven BioProjects known to mix lung with many other organs (PRJEB51634, PRJNA1005589, PRJNA1179423, PRJNA1188170, PRJNA1215450, PRJNA657844, and PRJNA902813), dropped accessions that lacked a BioProject accession, and dropped studies whose combined cell count was 1,000 or fewer. We sequentially fetched datasets that remained after these filters from GCS and computed their MD5 hashes on disk. Due to the extensive cumulative size of the datasets, we uploaded the datasets themselves, along with the newly computed hashes, to an internally owned and maintained Cloudflare R2 bucket, serving as a production mirror of scBaseCount, allowing us to remove local files from disk.

Atlas construction and quality control

To integrate the aquired individual datasets as a single atlas-scale `h5ad` file, we built a pipeline to sequentially download scBaseCount datasets from the R2 mirror, filter them by performing quality control operations, determine if what remained of the filtered datasets was sufficient for inclusion in the atlas, align the datasets to a fixed gene axis, and concatenate the datasets to one in-memory atlas. We wrote the concatenated atlas to disk on pipeline completion.

To ensure datasests matched scBaseCount, we computed dataset MD5 hashes following download from R2 and compared them the MD5 hashes stored as metadata alongside the corresponding R2 objects, the latter having been computed upon upload. We enforced identity between the upload and download hashes.

Next, we filtered cells based on tunable parameters. We did not drop genes for rarity. We flagged genes as likely mitochondrial, ribosomal, and hemoglobin genes by applying regular expressions to the gene names, and computed per-cell quality metrics. We first dropped cells with a total gene count below 200 genes. We dropped cells whose mitochondrial and ribosomal gene counts were at least 20% and 50%, respectively, of total gene counts. We set the hemoglobin fraction ceiling to 100%. No cells met that threshold, so this filter removed no cells. Following these filters, we applied a cutoff on number of remaining cells and percentage of cells remaining relative to the initial number of cells in the dataset. We did not include datasets in the atlas if they had fewer than 100 cells remaining or if fewer than 50% of cells remained in the dataset following filtering. We also excluded datasets in which every cell lacked a usable CZ CELLxGENE `cell_type` annotation (CZI Cell Science Program et al., 2025). We configured every mentioned threshold as a tunable parameter. To ensure that the values we settled on were well calibrated, we recorded the number of cells dropped by each filter for every file that reached QC, and we stored those counts both for all QC-processed files and for the subset of files that were concatenated, allowing us to rerun the filtering pipeline with different filtering parameters. We did not apply doublet detection. We chose filtering parameters based on recommendataions proposed by Luecken and Theis (2019) and enriched them with the ribosomal filter and hemoglobin flagging. Before we concatenated the filtered datasets along the cell axis to the atlas `h5ad` file, we reindexed each dataset onto the scBaseCount `geneInfo.tab` gene list, discarding genes absent from that list and filling missing reference genes with zeros, to ensure concatenated matrices used a shared gene axis. Since Youngblut et al. (2025) already indexed scBaseCount matrices against a shared STAR reference, this reindex did not change the genome reference.

Embedding and batch correction

We developed a method to embed and cluster the atlas using the CZ CELLxGENE `cell_type` key present in the datasets. First, we normalized gene counts per cell so that each cell had the same total gene count. Then, we logarithmized the data matrix by the natural logarithm, selected the top 2,000 highly variable genes (HVG) stratified by BioProject—a number in the range of recommended HVGs according to Luecken and Theis (2019)—and z-scaled the expression of each gene. We then applied principal component analysis

(PCA), computing the first 50 PCs. We recorded the minimum number of PCs in the range of 15–50 needed to explain at least 50% of the data's cumulative variance, or retained all 50 computed PCs when that target was not reached. We applied the processing steps through PCA using Scanpy (Wolf et al., 2018).

Following PCA, we applied Harmony to the PCs to correct for one batch effect (Korsunsky et al., 2019). The scBaseCount release from which we collected data did not contain any keys coarser than SRA experiment accession, which was effectively a one-to-one file key. We identified the NCBI BioProject accession as a key to use for applying batch correction to the atlas. BioProject accessions group SRA experiment accessions by project type, project scope, technical methodology, or other groupings (Barrett et al., 2012). To collect the BioProject accession for a given SRA experiment accession, along with other relevant metadata, we queried the European Nucleotide Archive's (ENA) portal API (Amid et al., 2020). As a result of the International Nucleotide Sequence Database Collaboration (INSDC), ENA and NCBI both host BioProject accessions (Arita et al., 2021).

To vizualise and cluster the data, we computed the nearest neighbors distance matrix and a neighborhood graph of observations and applied Scanpy's reproducible, single threaded UMAP dimensionality reduction implementation, which the atlas pipeline uses by default (McInnes et al., 2018). We clustered the neighborhood graph using the Leiden algorithm (Traag et al., 2019). To select an optimal Leiden clustering resolution, we sequentially applied Leiden clustering to the neighborhood graph at a range of resolutions from 0.1–1.9 at intervals of 0.1, computed the Jaccard index (also known as "intersection over union" or "IoU") of each CZ CELLxGENE `cell_type`-Leiden cluster pair, and determined the optimal one-to-one matching using a modified Jonker–Volgenant algorithm as implemented by SciPy's `linear_sum_assignment` function (Crouse, 2016; Virtanen et al., 2020). Because the function minimizes total assignment cost, we supplied the negated Jaccard matrix and summed the original Jaccard indices at the returned matches. We retained the Leiden clustering resolution with the highest matched Jaccard sum as the canonical Leiden clustering resolution. We also developed a random forest procedure that used stratified out-of-fold confusion between Leiden clusters based on highly variable gene expression to merge clusters that were not reliably distinguishable, but we did not apply this procedure to the atlas due to time constraints.

To test our clustering methodology, we implemented it as a set of importable functions and applied them to datasets corresponding with single SRA experiment accessions before applying it to the atlas. We drew one stratified sample of 100,000 cells for calibration and a separate stratified sample of 100,000 cells for validation. The validation sample was drawn using a different random seed from the calibration sample. For both samples, the strata were BioProjects, and the number of cells drawn from each stratum was proportional to that BioProject's share of the full atlas, with at least one cell per BioProject. We benchmarked the batch key using scIB (Luecken et al., 2022).

Results and discussion

There were 35,266 datasets available in the `2026-01-12` release of scBaseCount. We discarded 472 accessions whose identifiers were not SRA or ENA records. We retained accessions whose free-text tissue label matched a lung-related regular expression, leaving 2,396 accessions with lung-matched metadata. After retrieving study accessions from the ENA portal API, we removed 580 accessions that lacked a BioProject accession, belonged to one of the seven excluded multi-organ studies, or belonged to studies with a combined cell count of 1,000 or fewer. The remaining 1,816 accessions (13,013,643 barcodes; mean 7,166, median 5,112 barcodes per accession) formed the eligible set.

The atlas contained 1,410 datasets from 1,816 eligible datasets, spanning 172 BioProject accessions. All 1,816 eligible datasets reached per-cell QC (13,013,643 barcodes). After those cell filters, 10,893,794 barcodes remained. File-level gates then excluded 406 datasets, and concatenation retained 9,307,963 barcodes (Table 1). Restricted to the 1,410 concatenated datasets, the same cell filters removed 578,434 of 9,886,397 barcodes (145,320 below 200 genes, 430,729 at or above 20% mitochondrial counts, and 2,385 at or above 50% ribosomal counts). The atlas contained 61 CZ CELLxGENE `cell_type` labels (Fig 1).

Table 1. Atlas contained 1,410 datasets (9,307,963 barcodes) across 172 BioProject accessions, of 1,816 eligible datasets (13,013,643 barcodes) following filtering.

Filter	Datasets dropped, n (%)	Barcodes dropped, n (%)
MD5 hash (mismatch)	0 (0)	0 (0)
CZ CELLxGENE `cell_type` (labels absent)	238 (13.1)	1,509,128 (11.6)
Min. genes per cell ($<$ 200)	—	1,600,056 (12.3)
Max. mitochondrial fraction ($\geq$ 20%)	—	517,018 (4.0)
Max. ribosomal fraction ($\geq$ 50%)	—	2,775 (0.02)
Min. cells remaining ($<$ 100)	100 (5.5)	1,169 (0.01)
Min. fraction of cells remaining ($<$ 50%)	68 (3.7)	75,534 (0.58)

Dataset percentages use the 1,816 eligible datasets as the denominator. Barcode percentages use the 13,013,643 barcodes present in those datasets before QC. Per-cell filter counts include every file that reached QC, including files later rejected by file-level gates; they are not restricted to concatenated files. A total of 406 datasets were dropped. Of those, 0 were dropped due to MD5 hash mismatch, 238 (13.1% total datasets) were dropped due to fully absent CZ CELLxGENE `cell_type` labels. Of the 3,705,680 barcodes that were dropped from the eligible set, 1,600,056 (12.3% total barcodes) were dropped for gene counts of less than 200, 517,018 (4.0% total barcodes) were dropped for having 20% or more of total gene count being mitochondrial, and 2,775 (0.02% total barcodes) for having 50% or more of total gene counts being ribosomal. Following cell filtering, 100 (5.5% total datasets) were dropped for retaining fewer than 100 cells, and 68 (3.7% total datasets) were dropped for retaining less than 50% of cells. Among concatenated files only, cell filters dropped 578,434 barcodes.

On the 100,000-cell atlas validation subset, Harmony correction using NCBI BioProject as the batch key

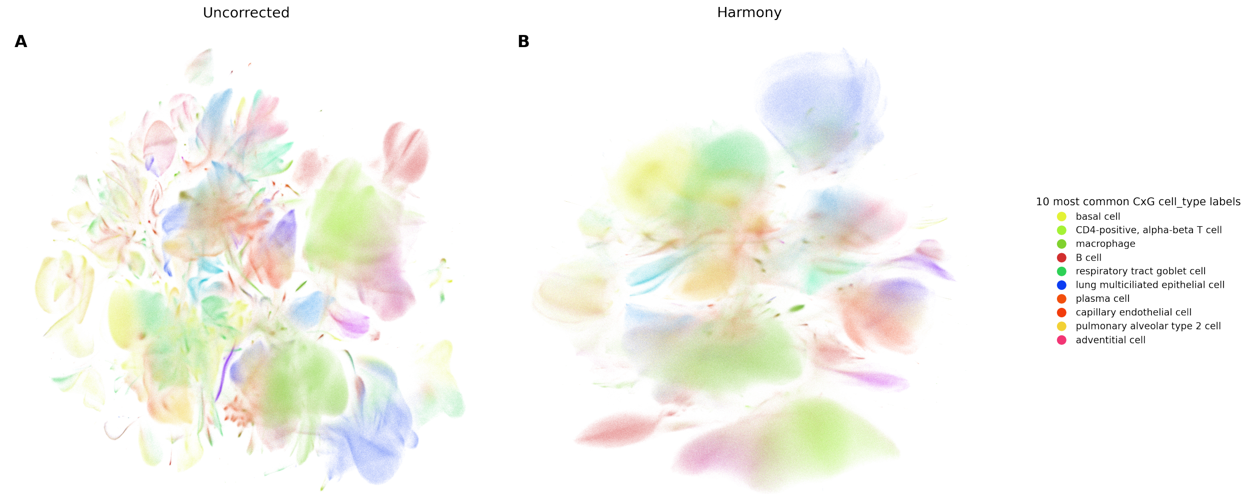

Fig 1. Single-file integrated lung disease atlas derived from scBaseCount. UMAP of the integrated atlas colored by CZ CELLxGENE `cell_type`. A: Uncorrected embedding. B: Harmony-corrected embedding (batch key: BioProject).

increased the aggregate batch correction score from 0.29 to 0.45, while the aggregate biological conservation score decreased from 0.52 to 0.50 (Fig 2A, Luecken et al. (2022)). Each aggregate is the unweighted mean of its five component metrics. Four of five individual scIB batch correction metrics increased, while one (graph connectivity) decreased from 0.61 to 0.47. All five scIB biological conservation metrics decreased, although their aggregate changed by only 0.02 (Fig 2B).

Harmony takes the PC coordinates of cells and attempts to correct them by moving cells to favor mixed representation of cells in clusters, while maintaining biological groupings (Korsunsky et al., 2019). Therefore, it is not surprising that we observed a gain of large magnitude on batch correction metrics, since the evaluation key is the same BioProject column used for correction. That gain is therefore not independent evidence that BioProject is the uniquely correct batch definition. However, cell-type conservation, which is scored against CZ CELLxGENE `cell_type`, exhibited minimal change. Other atlas collections can of course yield a different balance. Independent benchmarks place Harmony among methods that perform similarly to single-cell foundation models such as scGPT and Geneformer on batch correction, and indeed more stable than foundation models when they are used zero-shot, or without fine-tuning for batch correction (Cui et al., 2024; Kedzierska et al., 2025; Liu et al., 2026; Theodoris et al., 2023).

We validated our resolution selection methodology on five experiment files reproducibly sampled from the same metadata-selected cohort. Their empirical cell-count percentile ranks in the final eligible set were 55.62%, 15.01%, 75.17%, 16.02%, and 41.71% (2,072–9,102 cells before filtering; S1 Fig). On each file we ran Leiden clustering at resolutions 0.1–1.9 in steps of 0.1, scored each partition by the summed Jaccard index of

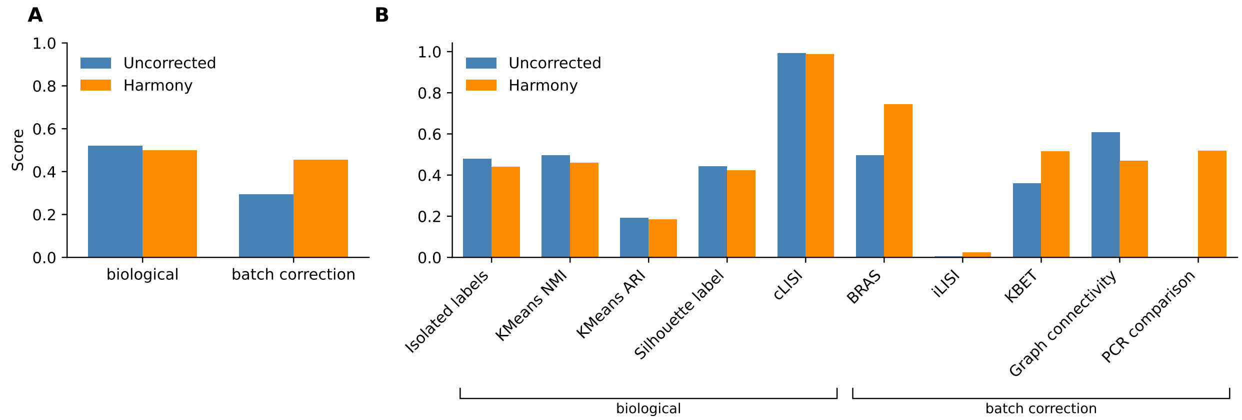

Fig 2. Batch correction metrics before and after Harmonization of PCs. A: Aggregate biological conservation and batch correction scores, each computed as the unweighted mean of its five component metrics. B: Individual scIB metrics. Uncorrected PCs versus Harmony-corrected PCs (batch key: NCBI BioProject). Aggregate biological conservation decreased from 0.52 to 0.50; aggregate batch correction increased from 0.29 to 0.45.

the optimal one-to-one matching between clusters and CZ CELLxGENE `cell_type` labels, and retained the resolution with the highest score. Selected resolutions ranged from 0.1 to 0.9, and the random-forest-merged partitions contained 4–19 clusters. On a 100,000-cell, study-stratified subset of the atlas the same matching criterion favoured resolution 0.8 (21 clusters in the subset). We used that resolution for the Harmony-corrected atlas, which produced 74 clusters.

Continued study is warranted using the present atlas. Due to the atlas spanning over 100 BioProject accessions, we recognize that differential gene expression analyses on various clusters has the potential to reaffirm disease state gene expression in certain cell types, or to highlight novel, low signal relationships that have the potential to represent new findings. As evident in the UMAP projection of the atlas, certain CZ CELLxGENE `cell_type` labels tend to cluster naturally, such as B and T lymphocytes (Fig 1 B). Sant et al. (2025) note that a cluster's ability to reliably representat real cellular identity and state is crucial for deriving biological meaning in downstream analysis steps. Since the clustering of the present atlas builds on the CZ CELLxGENE `cell_type` annotations as priors informing the clustering resolution, the clusters may indeed represent real biology.

Furthermore, we identified limitations with the study. Starting with atlas construction, reproduction requires cloud storage in the order of hundreds of GB. The concatenation of `h5ad` files was optimized for limited disk space rather than limited memory, and therefore requires that the entire concatenated atlas is held in memory before it is written to disk. Realistically, memory in the order of TB is required to safely reproduce the atlas or build a new one with the code present to avoid out-of-memory errors late in the concatenation. Additionally, given that scBaseCount is agentically curated, it is not surprising that we identified certain `h5ad` files that upon close inspection using the ENA portal API did not appear to entirely match the metadata perscribed by the agent built by Youngblut et al. (2025). While we sought out to identify reliably accurate fields in our deterministic context lookup on ENA via SRA experiment accession, it is entirely possible that certain metadata fields we rely on for context lookup can be missing or inaccurate, which can negatively affect downstream analysis, which relies on assumptions about which cells are in which state. Furthermore, there is potential to deepen the methodology of data processing. This is particularly true for the selection of HVGs. Multiple methods exist for ranking genes, and the number of genes plays an important role in the downstream granularity of the cell embedding, as well as with the materialization of batch effects (Kiselev et al., 2019; Li et al., 2026; Zhao et al., 2025). Neither HVG selection methodology nor HVG count were optimized in the methodology presented.

Conclusion

The field of single-cell technology continues to grow rapidly. With it, the availability of data, tools, and analysis methodology, has become difficult to navigate (Heumos et al., 2023). Meanwhile, expanded cell counts in scRNA-seq datasets is crucial for executing analysis of complex cell populations with rare cell types and states (Kolodziejczyk et al., 2015). Nonetheless, simply increasing the number of cells considered in scRNA-seq data analysis is not sufficient for generating those large datasets (Mueller et al., 2026). To produce processed datasets at large scale represents a challenge spanning data aquisition, processing, and correction for batch effects that materialize at atlas-level. Here, we propose a method for identifying and utilizing publically available resources in the form of large raw matrix stores—in this case scBaseCount—and integrating data to produce a single, analysis-ready atlas with extensive processing applied (Youngblut et al., 2025). The atlas supplies usable clusters, which the source files lacked.

The atlas building pipeline proposed here does not exist in isolation. During the writing of the present report, a Snakemake workflow for concatenating raw count matrices, and applying quality control and batch correction was developed by Mueller et al. (2026), a method that addresses the growing demand for curated atlases to be used for downlstream data analysis. Continued work would certainly be warranted in implementing the pipeline built by Mueller et al., 2026, as a means of validating the quality control and integration decisions made in the present method or producing more atlases.

Supporting information

S1 Fig. Single-SRX cluster validation for a reproducible random sample of five files from the eligible cohort. Pages 1–5, 6–10, 11–15, 16–20, and 21–25 show SRX9246208, SRX26487579, SRX24312779, SRX19727431, and SRX23395189, with empirical cell-count percentile ranks of 55.62%, 15.01%, 75.17%, 16.02%, and 41.71% in the final eligible set, respectively. Within each five-page block, page 1 shows cumulative PCA variance and the selected PC count. Page 2 shows cluster count and matched Jaccard score across resolutions 0.1–1.9; the horizontal grey line marks the number of retained CZ CELLxGENE `cell_type` labels. Page 3 shows UMAPs of the Jaccard-selected Leiden clustering, the random-forest-merged clustering, and the retained `cell_type` labels. Page 4 shows merged-cluster and `cell_type` composition. Page 5 shows homogeneity, completeness, V-measure, NMI, and ARI relative to the random-forest-merged clustering. The figure referenced can be found at https://github.com/otodreas/scBaseCount_Pipeline/blob/main/docs/report/si/resolution_validation.pdf.

Data and code availability

The code used to aquire, process and analyze data is tracked with Git and hosted on GitHub at https://github.com/otodreas/scBaseCount_Pipeline. Instructions to reproduce the results shown in this report can be found in the `README.md` file at the project root. The five validation `h5ad` files used for S1 Fig are distributed through Git LFS. At the time of submission (November 2026), we maintain a Cloudflare R2 bucket containing relevant `h5ad` files from scBaseCount, as well as build artifacts such as the quality controlled and concatenated atlas, as well as the final, post-processed atlas.

Usage of generative artificial intelligence

Nygen AB uses Cursor, an integrated development environment with features built around large language models (LLMs), across the company. We used Cursor to draft code and brainstorm code architecture. We did not use Cursor to write reports or analyze and interpret data. The repository presented here represents our own work. More information can be found at https://github.com/otodreas/scBaseCount_Pipeline/blob/main/README.md#generative-ai-usage.

References

Amid, C., Alako, B. T. F., Balavenkataraman Kadhirvelu, V., Burdett, T., Burgin, J., Fan, J., Harrison, P. W., Holt, S., Hussein, A., Ivanov, E., Jayathilaka, S., Kay, S., Keane, T., Leinonen, R., Liu, X., Martinez-Villacorta, J., Milano, A., Pakseresht, A., Rahman, N., … Cochrane, G. (2020). The European Nucleotide Archive in 2019. *Nucleic Acids Research*, *48*(D1), D70–D76. https://doi.org/10.1093/nar/gkz1063

Arita, M., Karsch-Mizrachi, I., Cochrane, G., & on behalf of the International Nucleotide Sequence Database Collaboration. (2021). The international nucleotide sequence database collaboration. *Nucleic Acids Research*, *49*(D1), D121–D124. https://doi.org/10.1093/nar/gkaa967

Barrett, T., Clark, K., Gevorgyan, R., Gorelenkov, V., Gribov, E., Karsch-Mizrachi, I., Kimelman, M., Pruitt, K. D., Resenchuk, S., Tatusova, T., Yaschenko, E., & Ostell, J. (2012). BioProject and BioSample databases at NCBI: Facilitating capture and organization of metadata. *Nucleic Acids Research*, *40*(D1), D57–D63. https://doi.org/10.1093/nar/gkr1163

Chakraborty, A., Mastalerz, M., Ansari, M., Schiller, H. B., & Staab-Weijnitz, C. A. (2022). Emerging Roles of Airway Epithelial Cells in Idiopathic Pulmonary Fibrosis. *Cells*, *11*(6), 1050. https://doi.org/10.3390/cells11061050

Crouse, D. F. (2016). On implementing 2D rectangular assignment algorithms. *IEEE Transactions on Aerospace and Electronic Systems*, *52*(4), 1679–1696. https://doi.org/10.1109/TAES.2016.140952

Cui, H., Wang, C., Maan, H., Pang, K., Luo, F., Duan, N., & Wang, B. (2024). scGPT: Toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, *21*(8), 1470–1480. https://doi.org/10.1038/s41592-024-02201-0

CZI Cell Science Program, Abdulla, S., Aevermann, B., Assis, P., Badajoz, S., Bell, S. M., Bezzi, E., Cakir, B., Chaffer, J., Chambers, S., Cherry, J. M., Chi, T., Chien, J., Dorman, L., Garcia-Nieto, P., Gloria, N., Hastie, M., Hegeman, D., Hilton, J., … Carr, A. (2025). CZ CELLxGENE Discover: A single-cell data platform for scalable exploration, analysis and modeling of aggregated data. *Nucleic Acids Research*, *53*(D1), D886–D900. https://doi.org/10.1093/nar/gkae1142

Heumos, L., Schaar, A. C., Lance, C., Litinetskaya, A., Drost, F., Zappia, L., Lücken, M. D., Strobl, D. C., Henao, J., Curion, F., Schiller, H. B., & Theis, F. J. (2023). Best practices for single-cell analysis across modalities. *Nature Reviews Genetics*, *24*(8), 550–572. https://doi.org/10.1038/s41576-023-00586-w

Kedzierska, K. Z., Crawford, L., Amini, A. P., & Lu, A. X. (2025). Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biology*, *26*(1), 101. https://doi.org/10.1186/s13059-025-03574-x

Kiselev, V. Y., Andrews, T. S., & Hemberg, M. (2019). Challenges in unsupervised clustering of single-cell RNA-seq data. *Nature Reviews Genetics*, *20*(5), 273–282. https://doi.org/10.1038/s41576-018-0088-9

Kolodziejczyk, A. A., Kim, J. K., Svensson, V., Marioni, J. C., & Teichmann, S. A. (2015). The Technology and Biology of Single-Cell RNA Sequencing. *Molecular Cell*, *58*(4), 610–620. https://doi.org/10.1016/j.molcel.2015.04.005

Korsunsky, I., Millard, N., Fan, J., Slowikowski, K., Zhang, F., Wei, K., Baglaenko, Y., Brenner, M., Loh, P.-r., & Raychaudhuri, S. (2019). Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, *16*(12), 1289–1296. https://doi.org/10.1038/s41592-019-0619-0

Leinonen, R., Sugawara, H., Shumway, M., & on behalf of the International Nucleotide Sequence Database Collaboration. (2011). The Sequence Read Archive. *Nucleic Acids Research*, *39*, D19–D21. https://doi.org/10.1093/nar/gkq1019

Li, S., Lücken, M., Marioni, J. C., Teichmann, S. A., & He, P. (2026). Toward informed batch correction for single-cell transcriptome integration. *Nature Computational Science*, *6*(2), 123–133. https://doi.org/10.1038/s43588-025-00943-1

Liu, T., Li, K., Wang, Y., Li, H., & Zhao, H. (2026). Evaluating the Utilities of Foundation Models in Single-Cell Data Analysis. *Advanced Science*, *13*(27), e14490. https://doi.org/10.1002/advs.202514490

Luecken, M. D., Büttner, M., Chaichoompu, K., Danese, A., Interlandi, M., Mueller, M. F., Strobl, D. C., Zappia, L., Dugas, M., Colomé-Tatché, M., & Theis, F. J. (2022). Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, *19*(1), 41–50. https://doi.org/10.1038/s41592-021-01336-8

Luecken, M. D., & Theis, F. J. (2019). Current best practices in single-cell RNA-seq analysis: A tutorial. *Molecular Systems Biology*, *15*(6), MSB188746. https://doi.org/10.15252/msb.20188746

McInnes, L., Healy, J., Saul, N., & Großberger, L. (2018). UMAP: Uniform Manifold Approximation and Projection. *Journal of Open Source Software*, *3*(29), 861. https://doi.org/10.21105/joss.00861

Mueller, M. F., Cujba, A.-M., Romanovskaia, D., Cohen, C. J., Bright, C. A., Lance, C., Ramirez-Suastegui, C., Strobl, D. C., Yuan, H., Hulsen, J., Naas, J., Limbeck, K., Kock, K. H., Halle, L., Knoll, R., Kfuri-Rubens, R., Aguilar-Fernandez, S., Parikh, S., Shitov, V. A., . . .

Luecken, M. D. (2026, August 2). *Building optimized single-cell reference atlases with scAtlasTb.* https://doi.org/10.64898/2026.07.30.741695

Navin, N., Kendall, J., Troge, J., Andrews, P., Rodgers, L., McIndoo, J., Cook, K., Stepansky, A., Levy, D., Esposito, D., Muthuswamy, L., Krasnitz, A., McCombie, W. R., Hicks, J., & Wigler, M. (2011). Tumour evolution inferred by single-cell sequencing. *Nature*, *472*(7341), 90–94. https://doi.org/10.1038/nature09807

Regev, A., Teichmann, S. A., Lander, E. S., Amit, I., Benoist, C., Birney, E., Bodenmiller, B., Campbell, P., Carninci, P., Clatworthy, M., Clevers, H., Deplancke, B., Dunham, I., Eberwine, J., Eils, R., Enard, W., Farmer, A., Fugger, L., Göttgens, B., … Yosef, N. (2017). The Human Cell Atlas. *eLife*, *6*, e27041. https://doi.org/10.7554/eLife.27041

Sant, C., Mucke, L., & Corces, M. R. (2025). CHOIR improves significance-based detection of cell types and states from single-cell data. *Nature Genetics*, *57*(5), 1309–1319. https://doi.org/10.1038/s41588-025-02148-8

Sikkema, L., Ramírez-Suástegui, C., Strobl, D. C., Gillett, T. E., Zappia, L., Madissoon, E., Markov, N. S., Zaragosi, L.-E., Ji, Y., Ansari, M., Arguel, M.-J., Apperloo, L., Banchero, M., Bécavin, C., Berg, M., Chichelnitskiy, E., Chung, M.-i., Collin, A., Gay, A. C. A., … Theis, F. J. (2023). An integrated cell atlas of the lung in health and disease. *Nature Medicine*, *29*(6), 1563–1577. https://doi.org/10.1038/s41591-023-02327-2

Sung, H., Filho, A. M., Laversanne, M., Ferlay, J., Siegel, R. L., Soerjomataram, I., Jemal, A., & Bray, F. (2026). Global cancer statistics 2024: GLOBOCAN estimates of incidence and mortality worldwide for 34 cancers in 186 countries. *Ca*, *76*(4), e70090. https://doi.org/10.3322/caac.70090

Theodoris, C. V., Xiao, L., Chopra, A., Chaffin, M. D., Al Sayed, Z. R., Hill, M. C., Mantineo, H., Brydon, E. M., Zeng, Z., Liu, X. S., & Ellinor, P. T. (2023). Transfer learning enables predictions in network biology. *Nature*, *618*(7965), 616–624. https://doi.org/10.1038/s41586-023-06139-9

Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From Louvain to Leiden: Guaranteeing well-connected communities. *Scientific Reports*, *9*(1), 5233. https://doi.org/10.1038/s41598-019-41695-z

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … van Mulbregt, P. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*(3), 261–272. https://doi.org/10.1038/s41592-019-0686-2

Wolf, F. A., Angerer, P., & Theis, F. J. (2018). SCANPY: Large-scale single-cell gene expression data analysis. *Genome Biology*, *19*(1), 15. https://doi.org/10.1186/s13059-017-1382-0

Youngblut, N. D., Carpenter, C., Prashar, J., Ricci-Tam, C., Ilango, R., Teyssier, N., Konermann, S., Hsu, P. D., Dobin, A., Burke, D. P., Goodarzi, H., & Roohani, Y. H. (2025, May 6). *scBaseCount: An AI agent-curated, uniformly processed, and continually expanding single cell data repository.* https://doi.org/10.1101/2025.02.27.640494

Zhao, R., Lu, J., Li, Y., Zhou, W., Zhao, N., & Ji, H. (2025). A systematic evaluation of highly variable gene selection methods for single-cell RNA-sequencing. *Genome Biology*, *26*(1), 424. https://doi.org/10.1186/s13059-025-03887-x